Synopsis

Project Title

End- to- End Machine Learning System for WiFi Congestion Prediction

Problem Id: 25

Student Name: BHARGAV ISTWAL

University Roll number: 44

Section: Q

- **Problem Description:**

In today's college environment, many users access Wi-Fi at the same time, which often leads to network congestion, high latency, and slow performance. This affects online classes and academic work. The main problem is that congestion is not predicted in advance, causing sudden network issues. This project aims to use machine learning to analyze network data such as number of devices, latency, bandwidth usage, and packet loss. By identifying patterns in this data, the system can predict whether the network will become congested or not, helping improve network management and provide a better user experience.

- **Steps of Implementation:**

1. The project starts by collecting Wi-Fi network data such as number of devices, latency, bandwidth usage, and packet loss.

2. The collected dataset is loaded into Python for processing.

3. The data is cleaned by removing incorrect values and handling missing data.

4. The dataset is normalized to ensure all features are on the same scale.

5. Exploratory Data Analysis (EDA) is performed using graphs to understand patterns in the data.

6. Important features affecting network congestion are identified and selected.

7. The dataset is divided into training and testing sets.

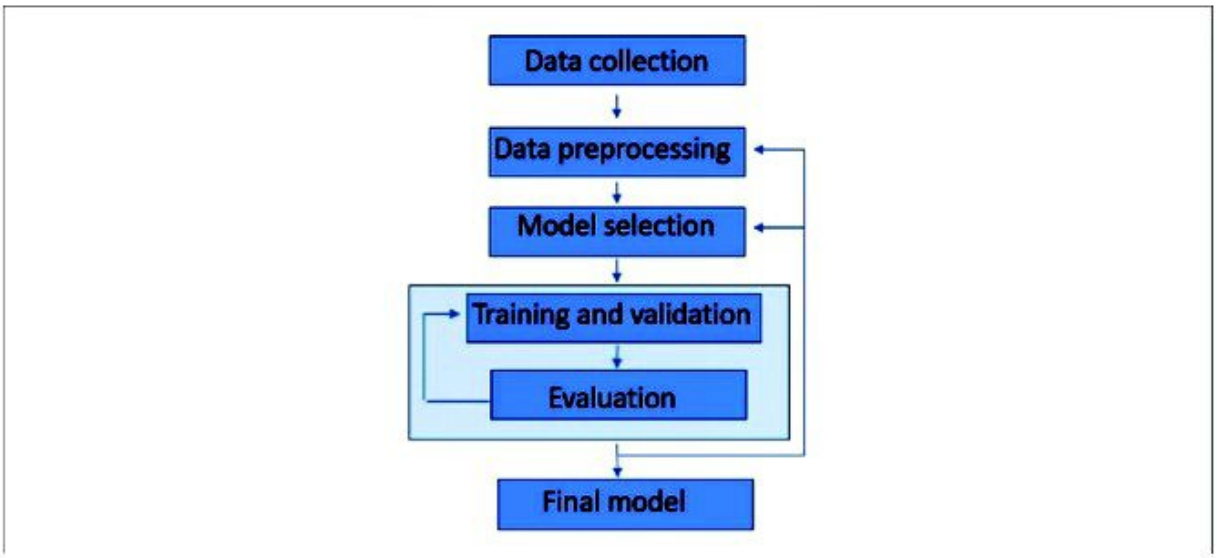

Fig. 1 Machine Learning Pipeline (Workflow Diagram)

Expected Modules/ Libraries to be used :

1. Pandas: Used for loading the dataset, handling missing values, and performing data manipulation operations on Wi-Fi network data.

2. NumPy: Used for numerical computations and working with arrays for efficient data processing.

3. Scikit-Learn (sklearn): Used for implementing the Logistic Regression model, splitting data, and preprocessing using tools like StandardScaler.

4. Matplotlib / Seaborn: Used for visualizing data through graphs such as latency trends, correlation plots, and performance charts.

5. Jupyter Notebook: Used as the development environment to write, run, and visualize the machine learning workflow step by step.

- **Expected Evaluation metrics and Plots to be used:**

1. Accuracy: Used to measure the overall percentage of correct predictions made by the model for detecting network congestion.

2. Precision and Recall: Precision ensures that non-congested networks are not wrongly classified as congested, while Recall ensures that actual congestion cases are correctly identified.

3. F1-Score: Used to balance Precision and Recall, especially when the dataset is not perfectly balanced.

4. Confusion Matrix: A table used to visualize correct and incorrect predictions, plotted using Matplotlib and Seaborn for better understanding.

5. ROC Curve: A graphical plot used to evaluate the model's ability to distinguish between congested and non-congested conditions, created using Matplotlib/Seaborn.

- **Required topics from the subject:**

1. Supervised Learning: It is a type of machine learning where the model is trained using labeled data. In this project, it helps in predicting whether the network is congested or not based on given input features.

2. Data Preprocessing: It involves cleaning and preparing the dataset by handling missing values, removing errors, and normalizing data. This step ensures better accuracy and performance of the model.

3. Feature Selection: It is used to select the most important features that affect network congestion. This helps in reducing complexity and improving model efficiency.

4. Model Training: In this step, the machine learning model is trained using the dataset so that it can learn patterns and relationships between input and output.

- 5. Model Evaluation: It involves checking the performance of the trained model using metrics like accuracy, precision, and recall to ensure correct predictions.

- **Platform Required:**

1. Jupyter Notebook / Google Colab: Used for writing and running Python code in an interactive environment. It helps in easy visualization of data and step-by-step execution of the machine learning process.

2. Python 3.x: The main programming language used for implementing the project. It provides libraries like Pandas, NumPy, and Scikit-learn for data analysis and machine learning.

3. Visual Studio Code / Python IDLE: Used as a coding platform for writing, editing, and managing Python scripts efficiently.

- **Books and Link Sources:**

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron: A practical guide for understanding machine learning concepts and implementing models using Python.

2. Introduction to Machine Learning with Python by Andreas C. Müller and Sarah Guido: Provides a clear explanation of basic ML algorithms and their real-world applications.

3. Kaggle (www.kaggle.com): An online platform used to find and download datasets related to network performance and data analysis.

4. Scikit-Learn Documentation (scikit-learn.org): Official documentation that provides guidance on implementing machine learning models like Logistic Regression and evaluation techniques.